

Dr. Doc:

#### Question 1

All of the following parts of the respiratory tract consist of hyaline cartilages, EXCEPT:

- A. Nasal septum
- B. Larynx
- C. Trachea
- D. Bronchi
- E. Bronchioles

Correct Answer: E. Bronchioles

Explanation:

- ◆ Bronchioles (E) are the small intralobular airways with diameters of 1 mm or less. Histologically, they are characterized by the complete absence of both cartilage and mucosal glands in their walls. Their structural patency is maintained instead by a prominent circular layer of smooth muscle.
- ◆ Nasal septum (A) is supported by bone and hyaline cartilage.
- ◆ Larynx (B) features a rigid wall reinforced by multiple cartilages, including hyaline cartilages (the thyroid, cricoid, and inferior arytenoid cartilages) and smaller elastic cartilages (such as the epiglottis).
- ◆ Trachea (C) contains a series of about a dozen C-shaped rings of hyaline cartilage in its submucosa to prevent lumen collapse during breathing.
- ◆ Bronchi (D) contain cartilage in their walls. While the primary bronchi have cartilage rings that completely encircle the lumen, these rings are gradually replaced by isolated plates of hyaline cartilage as the bronchial branches become smaller.

## Question 2

Indicate the wrong statement regarding the histology of the nasal cavity:

- A. The vestibule is lined anteriorly by skin containing thick, short hairs known as vibrissae.
- B. The nasal cavities proper consist of both olfactory and respiratory mucosae.
- C. The superior and the middle conchae are lined by olfactory mucosa.
- D. The inferior concha is lined by respiratory epithelium.
- E. Paranasal sinuses are lined by thin respiratory mucosa with fewer goblet cells.

Correct Answer: C. The superior and the middle conchae are lined by olfactory mucosa.

Explanation:

- ◆ Statement C is incorrect (wrong) because only the superior conchae and the roof of the nasal cavities are lined by specialized olfactory epithelium. The middle and inferior conchae are covered by typical respiratory epithelium.
- ◆ Statement A is correct: The external, dilated vestibule marks the entrance of the nasal cavity where the skin of the nose enters the nares. It contains sweat glands, sebaceous glands, and stiff, moist hairs called vibrissae that filter out large particulates.
- ◆ Statement B is correct: The nasal cavities proper contain two distinct types of mucosa: the specialized olfactory mucosa (for smell) and the respiratory mucosa (for conditioning air).
- ◆ Statement D is correct: The inferior conchae (along with the middle conchae) are lined by respiratory epithelium, which is a ciliated pseudostratified columnar epithelium.

◆ Statement E is correct: The paranasal sinuses are bilateral cavities in the skull bones lined by a thinner respiratory epithelium that has fewer goblet cells.

### Question 3

Indicate the wrong statement about the olfactory mucosa:

- A. It has limited distribution than respiratory mucosa.
- B. In humans, it is about 10 cm square area.
- C. It is thicker than that of the respiratory mucosa.
- D. Its epithelium is tall pseudostratified columnar with goblet cells.
- E. Its lamina propria is highly vascular (with many venous plexuses) and consists of Bowman's glands.

Correct Answer: D. Its epithelium is tall pseudostratified columnar with goblet cells.

Explanation:

✓Statement D is incorrect (wrong) because olfactory epithelium does NOT contain goblet cells. It is a specialized, thick pseudostratified columnar epithelium composed of only three primary cell types: bipolar olfactory neurons, columnar supporting (sustentacular) cells, and basal stem cells. Goblet cells are instead a hallmark of respiratory epithelium.

✓Statement A is correct: Olfactory mucosa has a restricted distribution limited to the roof of the nasal cavity and superior conchae, whereas respiratory mucosa lines the vast majority of the conducting system.

✓Statement B is correct: In adult humans, the olfactory epithelium is quite localized, covering a total surface area of approximately 10 cm<sup>2</sup>.

✓Statement C is correct: The olfactory epithelium is a highly developed, thick pseudostratified columnar epithelium measuring up to 100 µm in thickness, which is thicker than neighboring respiratory epithelium.

✓Statement E is correct: The lamina propria of the olfactory mucosa is highly vascularized and contains specialized tubuloalveolar serous glands called olfactory glands (of Bowman). These glands produce a constant flow of fluid to clean the olfactory cilia and solubilize new odorants.

#### Question 4

Bronchioles differ from intrapulmonary bronchi in that:

- A. The bronchiolar wall is free of cartilages and glands.
- B. Bronchioles are lined with respiratory epithelium that progressively turns to ciliated simple columnar in the terminal bronchioles.
- C. The terminal bronchioles contain non-ciliated cells known as Clara cells which are absent in the bronchi.
- D. The wall of respiratory bronchiole is interrupted by the openings of many alveoli, which are absent in bronchi.
- E. All of the above

Correct Answer: E. All of the above

Explanation:

All statements (A, B, C, and D) are accurate histological differences between bronchioles and intrapulmonary bronchi:

✓Statement A is correct: Intrapulmonary bronchi are supported by cartilage plates and contain seromucous glands. In contrast, bronchioles are completely free of cartilage and glands.

✓Statement B is correct: While larger bronchioles retain ciliated pseudostratified columnar epithelium, the lining gradually decreases in height and complexity as the airways branch, becoming simple ciliated columnar or simple cuboidal in the terminal bronchioles.

✓Statement C is correct: The terminal bronchioles contain dome-shaped, non-ciliated Clara cells (exocrine bronchiolar cells). These cells are highly protective—secreting surfactant lipoproteins, detoxifying xenobiotics, and acting as stem cells—and are completely absent in bronchi.

✓Statement D is correct: The wall of a respiratory bronchiole is structurally interrupted by the openings of alveoli (sites of gas exchange). Conducting bronchi and bronchioles do not feature alveolar openings.

#### Question 5

Which of the following connective tissue fibers found in the interalveolar septa contributes to the spongy nature of the lungs?

- A. Collagen fibers
- B. Elastic fibers
- C. Reticular fibers
- D. All of the above
- E. 'B' and 'C' only

Correct Answer: E. 'B' and 'C' only

Explanation:

✓Elastic fibers (B) and Reticular fibers (C) make up the primary sparse extracellular matrix of the thin interalveolar septa.

✓The elastic fibers are arranged to enable the alveoli to expand during inhalation and contract passively during exhalation.

✓The reticular fibers act as structural support, preventing both the collapse and excessive distention of the delicate alveoli.

✓Together, this fiber network within the septa is responsible for the characteristic spongy nature of the lung parenchyma. Classic thick collagen fibers (Type I) are not major structural components of the normal, thin interalveolar septa.

#### Question 6

The epithelial lining of a lobar bronchus is which of the following tissue types?

- A. Simple columnar
- B. Ciliated pseudostratified
- C. Stratified squamous
- D. Transitional
- E. Simple squamous

Correct Answer: B. Ciliated pseudostratified

Explanation:

- ◆ Ciliated pseudostratified columnar epithelium (commonly known as typical respiratory epithelium) lines the larger conducting airways of the respiratory tract, including the trachea and the bronchi.
- ◆ A lobar (secondary) bronchus is a major branch of the bronchial tree that supplies a pulmonary lobe.
- ◆ Like the primary bronchi, lobar bronchi are lined by typical ciliated pseudostratified columnar epithelium containing goblet cells, ciliated cells, and basal cells. It is only further down the airway (in

smaller bronchioles) that the epithelium gradually simplifies to simple columnar and simple cuboidal tissue.

#### Question 7

In a section of lung tissue consisting predominantly of alveoli, a tubule about 2 mm in diameter is observed that contains smooth muscle and cartilage in its wall. The tubule is which one of the following?

- A. Alveolar duct
- B. Alveolar sac
- C. Bronchiole
- D. Bronchus
- E. Trachea

Correct Answer: D. Bronchus

Explanation:

- ◆ Bronchi (D) branch repeatedly within the lungs, with their branches becoming progressively smaller until they reach a diameter of 1 to 2 mm. Even at these smaller dimensions, they still contain smooth muscle and plates of hyaline cartilage in their walls.
- ◆ Alveolar ducts (A) and alveolar sacs (B) are thin-walled respiratory structures lined by extremely attenuated simple squamous cells; they contain no cartilage plates.
- ◆ Bronchioles (C) are intralobular airways (typically 1 mm or less in diameter) that are histologically characterized by the complete absence of cartilage and mucosal glands in their walls.
- ◆ Trachea (E) is a much larger midline organ (10 to 12 cm long and over 1.5 cm in diameter) located outside the lungs; it would not appear as a 2 mm tubule surrounded by alveoli in a lung section.

## Question 8

Consider pulmonary surfactant. Which of the following is FALSE?

- A. It consists of a mixture of phospholipids, neutral lipids and proteins.
- B. Mostly secreted by type I pneumocytes.
- C. Coats the alveolar surface of the type I and type II pneumocytes.
- D. Some of its protein components come from Clara cells.
- E. Not secreted by fetal lung until after the 35th week of gestation.

Correct Answer: B. Mostly secreted by type I pneumocytes.

Explanation:

- ◆ Statement B is FALSE (making it the correct answer) because pulmonary surfactant is synthesized and secreted by Type II alveolar cells (type II pneumocytes) and Clara cells, NOT by Type I pneumocytes. Type I pneumocytes are extremely thin, attenuated cells specialized almost exclusively for gas exchange.
- ◆ Statement A is true: Surfactant is an oily mixture composed of lipids (primarily phospholipids like DPPC and cholesterol) and specific surfactant proteins.
- ◆ Statement C is true: Once extruded, the surfactant spreads over the entire inner alveolar surface as a film, coating both Type I and Type II pneumocytes to lower surface tension.
- ◆ Statement D is true: Clara cells in the terminal bronchioles contribute to the surfactant layer by secreting surfactant proteins and lipoproteins into the bronchiolar fluid.

◆ Statement E is true: In fetal development, surfactant only appears in the last weeks of gestation as Type II cells differentiate and form lamellar bodies. Premature infants born before this system matures suffer from infant respiratory distress syndrome due to a lack of adequate surfactant.

#### Question 9

Consider Clara cells. Which of the following is FALSE?

- A. Also known as bronchiolar cells.
- B. Line conducting bronchioles along with goblet cells.
- C. Have a characteristic dome-shaped apical end that projects into the lumen.
- D. Secrete some of the protein components of surfactant to aid in maintaining patency of the bronchiole.
- E. Serve as stem cells for the bronchiolar epithelium.

Correct Answer: B. Line conducting bronchioles along with goblet cells.

Explanation:

◆ Statement B is FALSE (making it the correct answer) because goblet cells and mucosal glands are completely absent in the bronchiolar wall. As the airways transition from bronchi to bronchioles, the goblet cells are replaced by non-ciliated Clara cells.

◆ Statement A is true: Clara cells are widely known as exocrine bronchiolar cells.

◆ Statement C is true: They have a very distinctive histological appearance, characterized by nonciliated, dome-shaped apical ends that bulge into the airway lumen.

◆ Statement D is true: Clara cells secrete components of surfactant (including lipoproteins and mucins) to reduce surface tension and prevent bronchiolar collapse during exhalation.

◆ Statement E is true: They include a vital stem cell subpopulation that undergoes mitosis to regenerate and replace other ciliated and non-ciliated cells within the bronchiolar epithelium after an injury.

#### Question 10

The true vocal folds of the larynx:

- A. Are normally covered by a maximally keratinized stratified squamous epithelium.
- B. Contain a smooth muscle known as the trachealis.
- C. Contain a smooth muscle known as the vocalis.
- D. Contain many seromucous glands.
- E. Contain the vocal ligament that is rich in elastic fibers.

Correct Answer: E. Contain the vocal ligament that is rich in elastic fibers.

Explanation:

◆ Statement E is correct: The free edge of each true vocal fold is supported internally by the vocal ligament, which consists of a dense regular bundle of elastic connective tissue.

◆ Statement A is incorrect: The vocal folds are covered by protective, nonkeratinized stratified squamous epithelium, which protects the mucosa from abrasion and desiccation during rapid air movements. Maximally keratinized epithelium belongs to the skin.

◆ Statement B is incorrect: The trachealis is a smooth muscle that bridges the posterior opening of the C-shaped cartilage rings in the trachea, not the larynx.

◆ Statement C is incorrect because the vocalis muscle consists of large bundles of striated skeletal muscle fibers, NOT smooth muscle.

◆ Statement D is incorrect: Unlike the surrounding laryngeal chambers and vestibular folds, the true vocal folds themselves do not contain seromucous glands in order to remain thin and mobile for phonation.

#### Question 11

Typical respiratory epithelium consists of all of the following cells, EXCEPT:

- A. Columnar ciliated cells
- B. Goblet cells
- C. Brush cells
- D. Hair cells
- E. Basal cells

Correct Answer: D. Hair cells

Explanation:

◆ Hair cells (D) are specialized neuroepithelial mechanoreceptors associated with the inner ear rather than the respiratory tract.

◆ Ciliated columnar cells (A) are the most abundant cell type in typical respiratory epithelium, featuring 250–300 cilia on their apical surfaces to move mucus.

◆ Goblet cells (B) are highly numerous mucous-secreting cells characterized by apical domains packed with mucin glycoprotein granules.

◆ Brush cells (C) represent about 3% of the epithelial population and act as chemosensory receptors with short, blunt microvilli.

- ◆ Basal cells (E) are small, mitotically active stem and progenitor cells situated on the basement membrane that continuously replenish the other epithelial cell types.

#### Question 12

By which of the following histological features, extrapulmonary bronchi differ from the trachea?

- A. By their lining epithelium
- B. By presence of discontinuous plates of cartilages in their wall
- C. By the abundance of the goblet cells in their epithelium
- D. By presence of numerous sero-mucous glands in their submucosa
- E. All of the above.

As it is written, there is no correct answer.

If we are comparing the trachea to intrapulmonary bronchi, the difference is the presence of discontinuous cartilage plates. If we are strictly comparing the trachea to extrapulmonary (primary) bronchi, the difference is that the primary bronchi have complete circular cartilage rings rather than C-shaped rings.

Here is how the trachea, extrapulmonary (primary) bronchi, and intrapulmonary bronchi actually differ:

- ◆ The Trachea: Has open, C-shaped rings of hyaline cartilage closed posteriorly by the trachealis muscle.
- ◆ Primary (Extrapulmonary) Bronchi: the segments outside the lungs. Histologically, they are nearly identical to the trachea. They do not have discontinuous plates of cartilage. The cartilage rings completely encircle the lumen (no longer just C-shaped).

Intrapulmonary Bronchi: As they enter the lung and the diameter decreases, the rings are replaced by isolated (discontinuous) plates of hyaline cartilage.

### Question 13

Which tissue of the bronchial wall is responsible for a marked constriction of bronchioles in a clinical condition known as 'asthma'?

- A. The bronchiolar lining epithelium
- B. The lamina propria
- C. The smooth muscle layer of the bronchioles
- D. The adventitia of the bronchioles
- E. None of the above

Correct Answer: C. The smooth muscle layer of the bronchioles

Explanation:

- ◆ Asthma is a chronic inflammatory disorder of the bronchial tree characterized by bronchospasms, which are sudden, severe constrictions of the smooth muscle in the bronchioles.
- ◆ While bronchi contain cartilage plates to help resist collapse, bronchioles completely lack cartilage and rely on a prominent circular layer of smooth muscle to maintain patency.
- ◆ Because the smooth muscle layer is proportionately much thicker in the bronchiolar wall than in the bronchi, its contraction has a dramatic, closing effect on the airway lumen during an asthma attack.

#### Question 14

A chronic lung disease characterized by enlargement of the air spaces distal to the bronchioles into large cyst-like sacs, with destruction of the interalveolar wall is:

- A. Asthma
- B. Emphysema
- C. Hyaline membrane disease
- D. Sinusitis
- E. Bronchitis

Correct Answer: B. Emphysema

Explanation:

- ◆ Emphysema (B) is a chronic lung disease (highly associated with cigarette smoking) characterized by the permanent dilation and enlargement of the respiratory air passages distal to the terminal bronchioles.
- ◆ This disease process involves the destruction of the delicate interalveolar walls (septa), converting clusters of small, functional alveoli into large, non-functional cyst-like sacs.
- ◆ This structural destruction causes an irreversible loss of alveolar surface area and pulmonary elastic fibers, severely impairing gas exchange and passive lung recoil.

#### Question 15

Which of the following cells cover the majority of the alveolar gas exchange surface?

- A. Pneumocytes type I
- B. Pneumocytes type II
- C. Alveolar macrophages
- D. Goblet cells
- E. None of these

Correct Answer: A. Pneumocytes type I

Explanation:

- ◆ Type I alveolar cells (type I pneumocytes) (A) are extremely thin, attenuated simple squamous epithelial cells.
- ◆ Even though Type II pneumocytes are more numerous in actual cell count, Type I cells are so broadly stretched out that they cover approximately 95% of the total alveolar surface area.
- ◆ These cells form the primary epithelial component of the ultra-thin blood-air barrier (respiratory membrane), which is optimized for rapid gas diffusion. Type II pneumocytes cover only the remaining 5% of the surface.

Question 16

The conducting zone components include:

- A. Epithelial lining
- B. Cartilage
- C. Smooth muscle
- D. All of the above

Correct Answer: D. All of the above

Explanation:

The conducting portion is structurally designed to clean, humidify, and warm inspired air while providing a secure conduit for air movement. Its wall is composed of a specialized combination of these tissues:

- ◆ Epithelial lining (A): Almost the entire conducting portion is lined with typical respiratory mucosa, which features ciliated pseudostratified columnar epithelium to trap and sweep away debris.
- ◆ Cartilage (B) & Smooth muscle (C): To prevent collapse and ensure an uninterrupted air supply, a combination of hyaline cartilage, elastic and collagen fibers, and smooth muscle provides the conducting airways with structural support, elasticity, and flexibility.

Question 17

The tracheal lining is characterized by all of the following EXCEPT:

- A. Pseudostratified ciliated columnar epithelium
- B. Goblet cells
- C. Simple squamous epithelium
- D. Basal cells

Correct Answer: C. Simple squamous epithelium

Explanation:

✓The trachea is lined by typical respiratory mucosa, which is a ciliated pseudostratified columnar epithelium. This lining includes several specialized cell types but completely lacks simple squamous epithelium.

✓Pseudostratified ciliated columnar epithelium (A) is the exact epithelial type that characterizes the tracheal lining.

✓Goblet cells (B) are highly abundant within this epithelial layer to secrete protective mucus.

✓Basal cells (D) sit on the basement membrane and serve as mitotically active stem and progenitor cells to replenish the other epithelial cells.

✓Simple squamous epithelium (C) is extremely thin and flat, making it highly specialized for gas exchange. Consequently, it is restricted to the respiratory portion (lining the alveoli as Type I pneumocytes) and capillary walls, and is never found lining the conducting trachea.

#### Question 18

Pneumocyte Type II deficiency in a newborn results in:

- A. Pulmonary hypertension
- B. Obstructive Sleep Apnea
- C. Emphysema
- D. Neonatal Respiratory Distress Syndrome (Hyaline Membrane Disease)

Correct Answer: D. Neonatal Respiratory Distress Syndrome (Hyaline Membrane Disease)

Explanation:

◆ Type II pneumocytes are responsible for synthesizing and secreting pulmonary surfactant, an oily mixture of phospholipids and proteins that spreads over the alveolar surface to lower surface tension.

◆ Surfactant prevents the collapse of alveoli during exhalation and eases the physical force required to expand them during inhalation.

- ◆ In premature infants, incomplete differentiation of Type II pneumocytes leads to a severe deficit of surfactant. Without enough surfactant, expanding the lungs becomes extremely difficult, resulting in Neonatal Respiratory Distress Syndrome (also known as Infant Respiratory Distress Syndrome or Hyaline Membrane Disease).

#### Question 19

When examining a histology slide, which feature is characteristic of a transverse section of a bronchiole (as opposed to a bronchus)?

- A. Presence of cartilage plates
- B. Pseudostratified ciliated columnar epithelium
- C. Absence of cartilage and presence of Clara (club) cells
- D. Extensive submucosal glands

Correct Answer: C. Absence of cartilage and presence of Clara (club) cells

Explanation:

- ◆ As the conducting system branches deeper into the lung, the airways undergo a major histological transition from bronchi to bronchioles:

- ◆ Absence of cartilage and glands: While bronchi contain irregular plates of hyaline cartilage and abundant seromucous glands in their walls, bronchioles completely lack cartilage and mucosal glands.

- ◆ Presence of Clara cells: Clara cells (exocrine bronchiolar cells) are highly protective, dome-shaped, non-ciliated secretory cells that are extremely numerous in the simple cuboidal to columnar epithelium of terminal bronchioles but are absent in bronchi.

- ◆ Therefore, the complete absence of cartilage and glands combined with the presence of Clara cells (C) is the definitive diagnostic feature distinguishing a bronchiole from a bronchus on a slide.